

Module	Parameters	Interpretable?
<i>Encoder Θ_{enc} — deep ML weights (opaque)</i>		
① Static GraphSAGE ($\times 2$ layers)	3,520	no
② Dynamic SAGE + GRU + multi-scale TCN + fuse	29,408	no
③ Gated fusion + V-head	98	no
<i>encoder subtotal</i>	<i>33,026</i>	
<i>Behavioural primitives — interpretable RUM coefficients</i>		
④ $\beta_{t,m}$ (24 hour $\times$ 3 mode time sensitivity)	72	yes
④ γ (distance decay)	1	yes
④ δ_k (occupational match per income tier)	2	yes
④ κ_k (income tier scaling, $\kappa_0 \equiv 1$)	2	yes
<i>RUM subtotal</i>	<i>77</i>	
Total trainable parameters	33,103	—
<i>Interpretable share</i>	<i>77 / 33,103 = 0.23%</i>	

Note: Stage 2 deployment introduces no new parameters; BPR coefficients ($\alpha = 0.15$, $b = 4$, multiplier cap = 2.5) are hyperparameters fixed per Sheffi (1985) and Beckmann et al. (1956).